

Montasr Mohamed

CRT

1.

- a) Four legal (allowed) names:

- totalScore
- max_value
- itemCount2
- isComplete

- b) Four illegal (not allowed) names and why:

- 2ndPlayer — Not allowed because names cannot start with a number.
- total-count — Not allowed because hyphens are not allowed.
- class — Not allowed because it is a special word Java already uses.
- my score — Not allowed because names cannot have spaces.

5.

- a) Primitive data type vs. abstract data type:

- A **primitive type** (like a basic number or true/false value) is built right into Java and holds just one simple value. An **abstract type** is a bigger custom type (like a String or a class) that can hold a lot of data and actions together.

- b) Class vs. object:

- A **class** is just the blueprint or plan. An **object** is the actual thing built from that plan that you can use in your program.

11.

Using `int j = 5; double k = 1.6; int y; double z;`:

- a) `y = j * k;`

- **New line:** `y = (int)(j * k);`

- **Why:** Multiplying an integer by a decimal gives a decimal answer. Because y can only hold whole numbers, you must force it to change into an integer using (int) so it drops the decimal.
- **b) $z = j * k;$**
 - **New line:** $z = j * k;$ (No change needed)
 - **Why:** The answer is already a decimal, and z is made to hold decimals, so Java handles it automatically.